

Chapter 5 Homework Answer Sheet

Part A Multiple choice (15 points)

1-5 BCEDD

6-10 DBDEA

11-15 CECAD

Part B Short questions (15 points)

16. (4 points)

(a) (2 pts) At constant pressure, $\Delta E = \Delta H - P\Delta V$. We must know the values of P and ΔV to calculate ΔE from ΔH .

(b) (2 pts) ΔE is larger than ΔH . Because there are 3 mole of gaseous reactant and no gaseous product, the quantity $(-P\Delta V)$ is positive. We add a positive quantity to ΔH to calculate ΔE , so ΔE must be larger.

17. (5 points)

(a) (1 pt) exothermic

(b) (2 pts) $q = (3.59 \text{ g}/24.30 \text{ g}\cdot\text{mol}^{-1}) \times (1204 \text{ kJ}/2\text{mol}) = 88.9 \text{ kJ}$

(c) (2 pts) $m = (234 \text{ kJ} \times 2\text{mol}/1204 \text{ kJ}) \times 40.30 \text{ g/mol} = 15.7 \text{ g}$

18. (3 points)

Equation: (4) $2 \text{ B(s)} + 3 \text{ H}_2\text{(g)} \rightarrow \text{B}_2\text{H}_6\text{(g)}$

(4) = (1)*1/2 + (2)* 3/2 - (3)

$\Delta H_4 = \Delta H_1*1/2 + \Delta H_2* 3/2 - \Delta H_3 = 35.4 \text{ kJ}$

19. (3 points)

$$0.1000 \text{ L} \times \frac{0.500 \text{ mol HCl}}{\text{L}} \times \frac{118 \text{ kJ heat released}}{2 \text{ mol HCl}} = 2.95 \text{ kJ of heat released if HCl limiting}$$

$$0.3000 \text{ L} \times \frac{0.100 \text{ mol Ba(OH)}_2}{\text{L}} \times \frac{118 \text{ kJ heat released}}{\text{mol Ba(OH)}_2} = 3.54 \text{ kJ heat released if Ba(OH)}_2 \text{ limiting}$$

Because the HCl reagent produces the smaller amount of heat released, HCl is limiting and 2.95 kJ of heat are released by this reaction.

$$\text{Heat gained by solution} = 2.95 \times 10^3 \text{ J} = \frac{4.18 \text{ J}}{^\circ\text{C g}} \times 400.0 \text{ g} \times \Delta T$$

$$\Delta T = 1.76^\circ\text{C} = T_f - T_i = T_f - 25.0^\circ\text{C}, T_f = 26.8^\circ\text{C}$$